

How Useful is LLM-Guided Reflection?

Evaluating the Impact of LLM-guided Reflection on Learning Outcomes with Interactive AI-Generated Educational Podcasts

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- Tutoring is incredibly powerful
- Reflection is the process of critically examining one's own learning and understanding
- Ex: *Please describe the most important things that you've learned so far, and what is your plan moving forward?* ¹
- Can LLM-guided reflection enhance learning outcomes?

¹<https://dl.acm.org/doi/10.1145/3448139.3448166>

Research Questions

- **RQ1:** Do interactive reflection prompts improve learning outcomes when incorporated into AI-generated podcasts?
- **RQ2:** Do interactive reflection prompts improve user experience when incorporated into AI-generated podcasts?

Background

- In prior work reflection was only evaluated at the end
- No one tested whether LLM-prompted reflection actually helps
- We need to test if LLM-prompted reflection actually helps

Technical Details

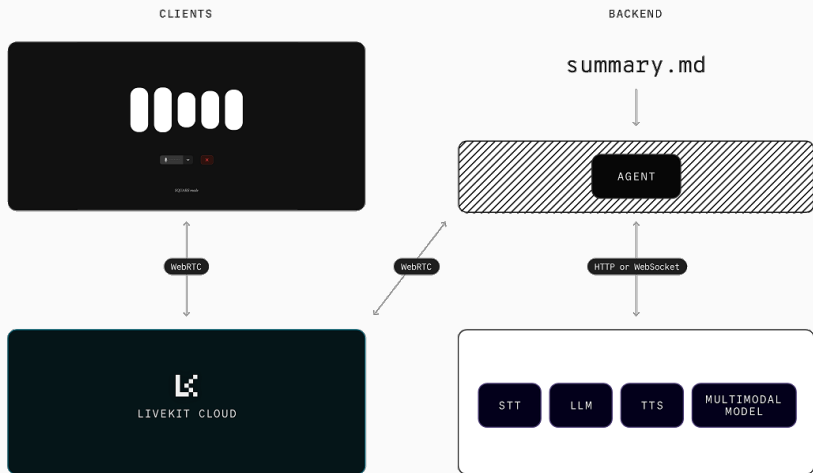


Figure 1: A diagram of our system, adapted from LiveKit's Agents

Study Design

- Web-based study with verbal AI podcast tutors
- 36 undergraduates, no previous philosophy background learning an Intro to Philosophy²
- Two modes: standard and reflection
- Sessions lasted 15-30 minutes
- At the end, participants were tested

²<https://openstax.org/books/introduction-philosophy/pages/1-1-what-is-philosophy>

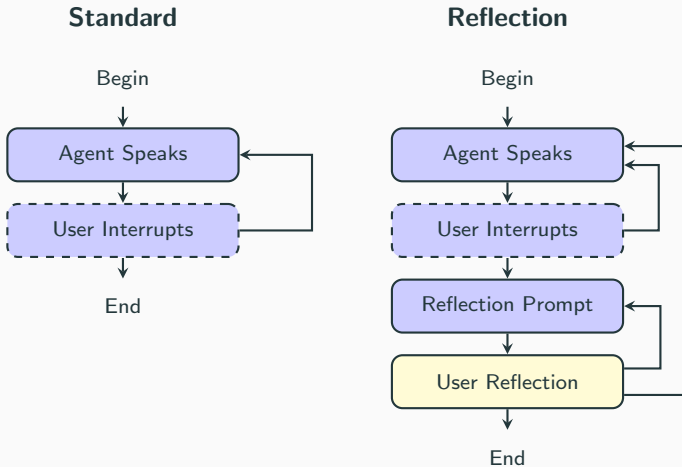


Figure 2: Standard vs Reflection Interaction Modes.

Hypothesis

We hypothesized that LLM-guided reflection will lead to

- Better learning outcomes
- Improved user experience
- Higher engagement

Procedure

- Participants accessed the web-based tutor and were taught on Intro to Philosophy
- They were allowed to interrupt and ask questions at any time
- In reflection they were given periodic prompts to reflect on learning
- User experience measured via UEQ (User Experience Questionnaire)

Our results ran counter to our theory, we found

- No difference in learning ($t=0.89$, $p=0.38$)
- Worse 'attractiveness' ratings ($t=2.26$, $p=0.03$, $D=0.75$)
- Users complained that LLM 'reflection' disrupted flow

Some Numbers

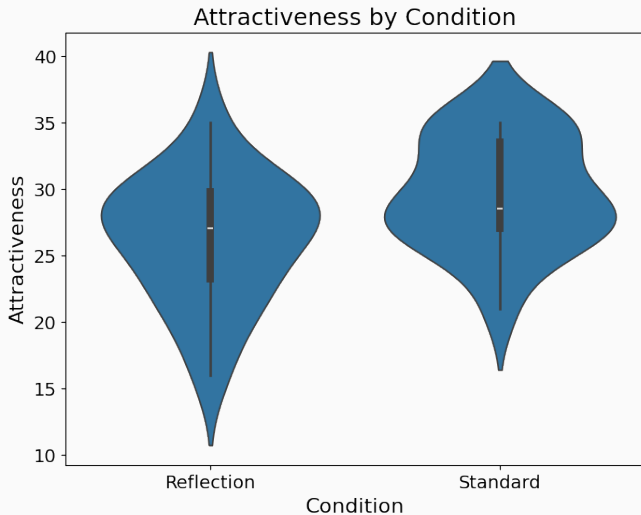


Figure 3: Attractiveness ratings across conditions.

Possible Explanations and Lessons

- Use more data: learning context, eye tracking, physiological signals
- Investigate when learners actually want or need to reflect
- Help learners build agency

Questions?

Thank You!
